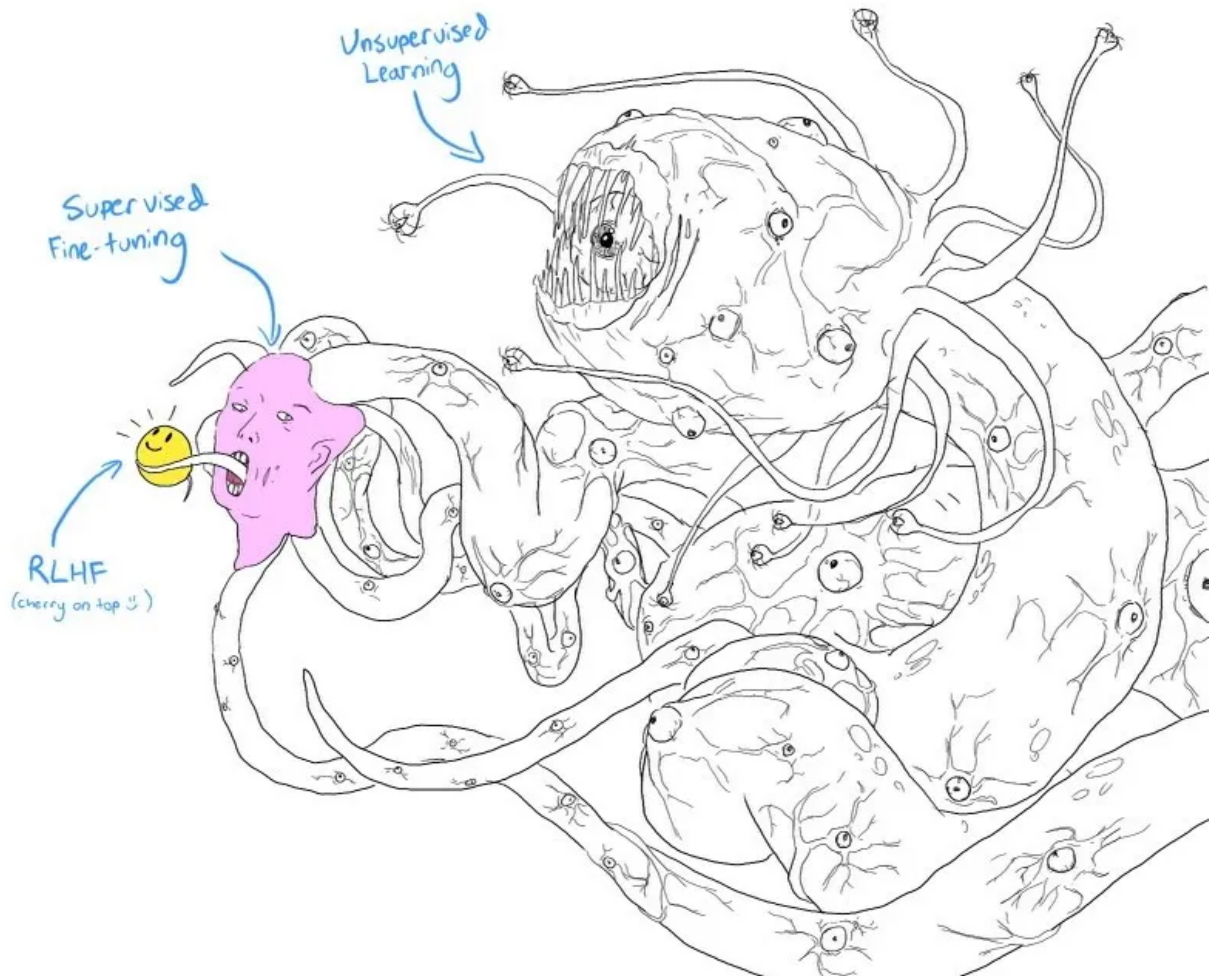
Reinforcement Learning

HSE, winter - spring 2026

Lecture 10: RL in a context of LLM



Sergei Laktionov

The full history of the meme is here

Agenda

1. LLM training pipeline
    1. Pretraining
    2. Instruction tuning (SFT)
    3. Alignment

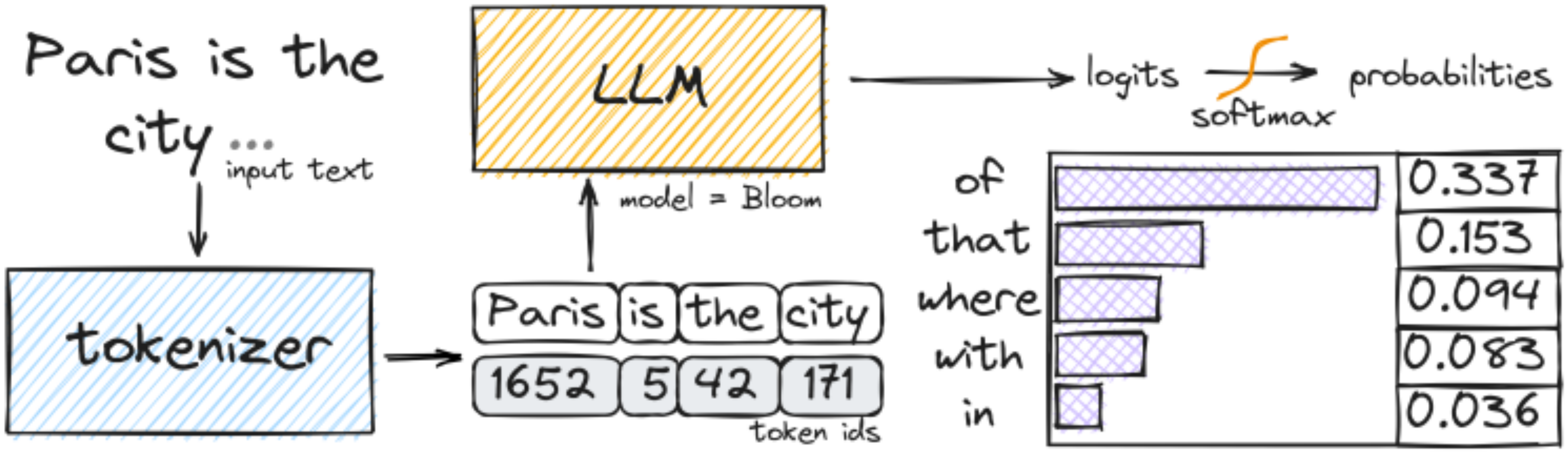
2. Deep dive into RL
    1. Reinforcement Learning from Human Feedback
    2. Reinforcement Learning from Verifiable Reward

Notation

- Prompt: x
- Generated sequence: $y = (y_1, \ldots, y_T)$
- LLM policy: $\pi_\theta(y \mid x)$

Autoregressive Policy

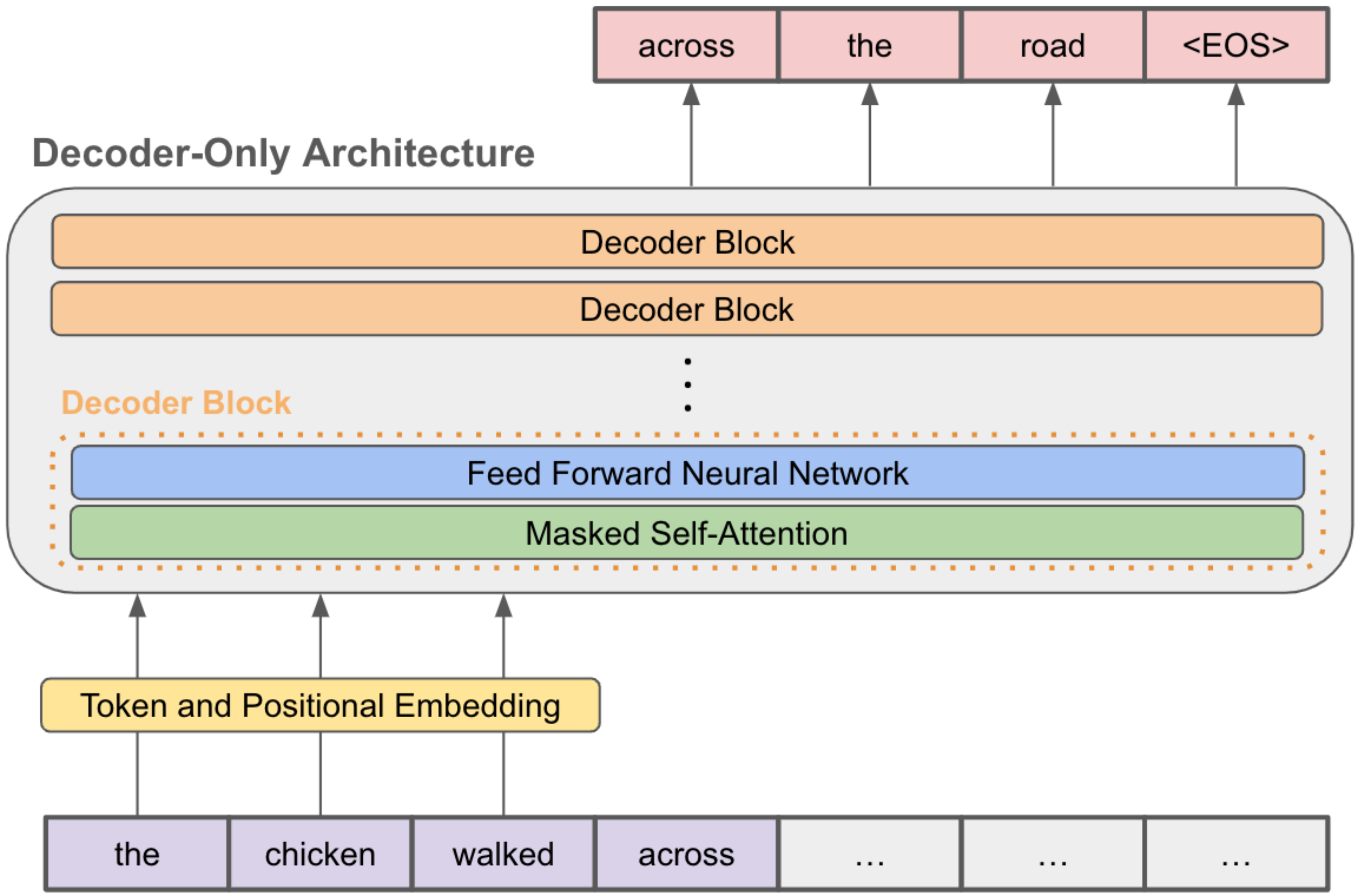
- Generation is sequential decision-making: $y_t \sim \pi_\theta(\,.\, | x, y_{<t})$
- Sequence probability factorization: $\pi_\theta(y \,|\, x) = \prod_{t=1}^{T} \pi_\theta(y_t \,|\, x, y_{<t})$



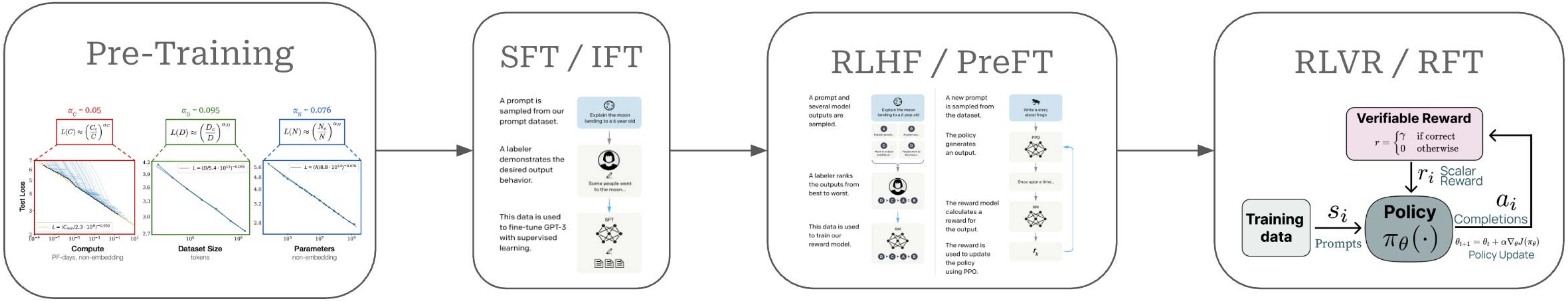
Source

Attention is All You Need

- $Attention(Q,K,V) = softmax(\frac{QK^T}{\sqrt{d}})V$
- Allows tokens to attend to each other
- Key mechanism enabling long-range dependencies

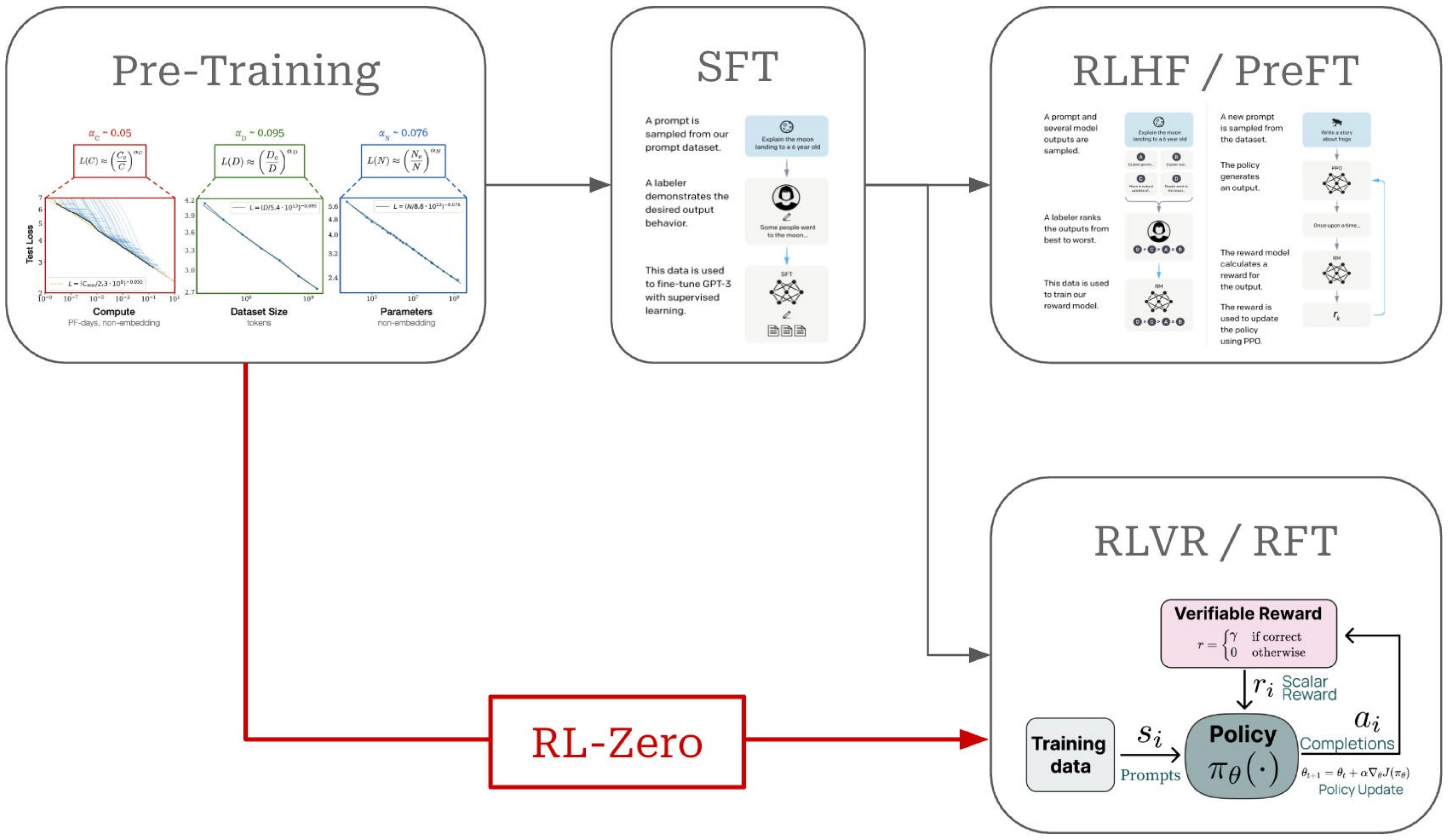


Training Stages



Source

Alternative Training Stages



[Source]

Pretraining Objective

- Maximum likelihood training: $\mathbb{E}_x[\sum_t \log \pi_\theta(x_t \mid x_{<t})] \to \max_\theta$
- Pretraining provides linguistic structure and knowledge, model approximates language distribution
- Datasets: $10^{11} - 10^{13}$ tokens

Instruction Fine-Tuning

- Dataset: (x, y), where x = prompt, y = golden answer
- Goal: teach models to follow instructions
- $\mathbb{E}_{(x,y)\sim\mathscr{D}} \log \pi_\theta(y \,|\, x) \rightarrow \max_\theta$

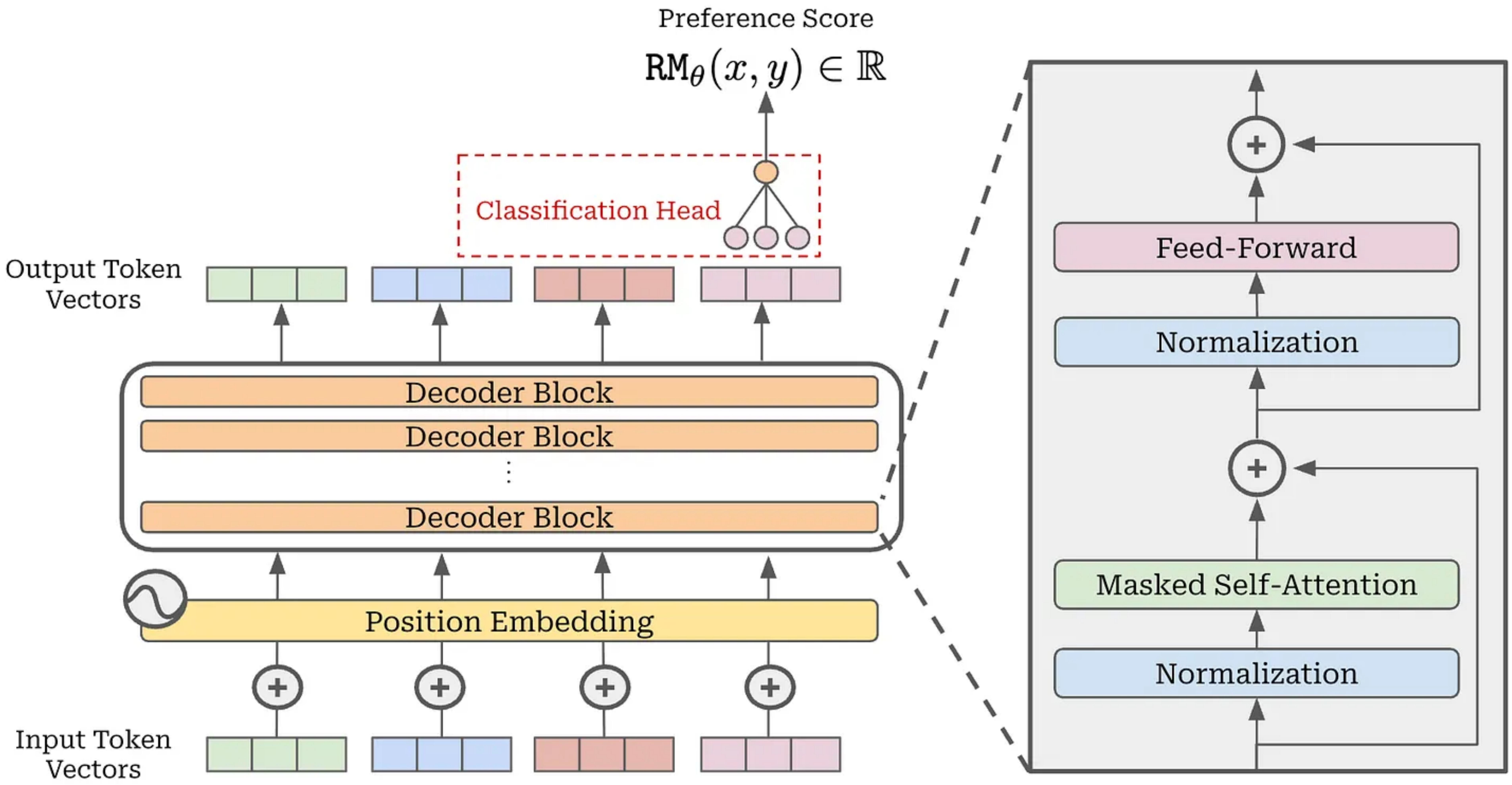
RLHF Motivation

- LLM is able to generate diverse and compelling text from human input prompts.
- What makes a "good" text is inherently hard to define as it is subjective and context dependent

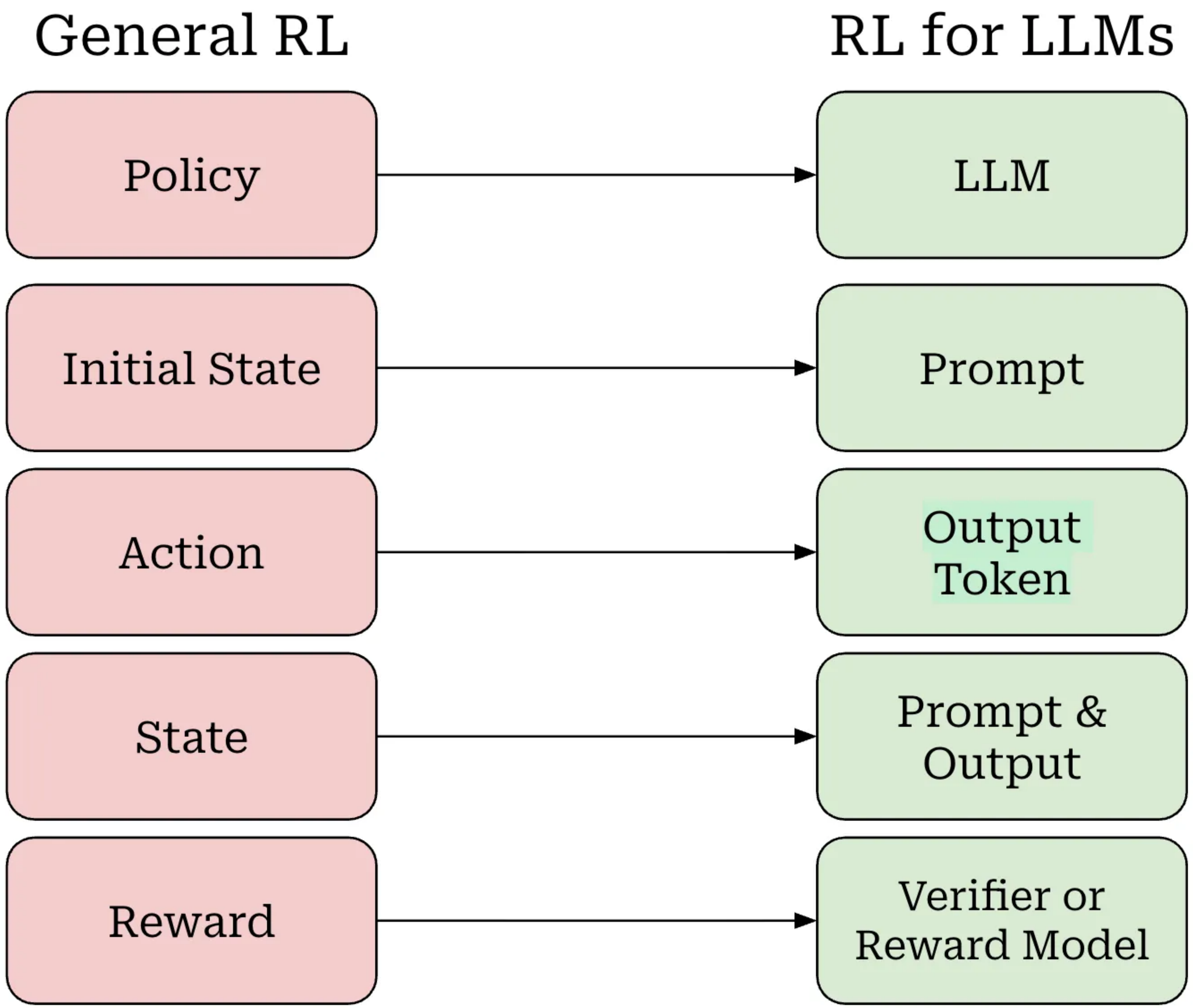
- Example: depending on the context the phrases:
  “I’m a robot” and “I am a robot” might be prefered differently even though the difference is marginal.

Reward Model

- $R_\phi(x, y)$ predicts preference score, i.e. approximates human evaluation.



RL Formulation for LLM



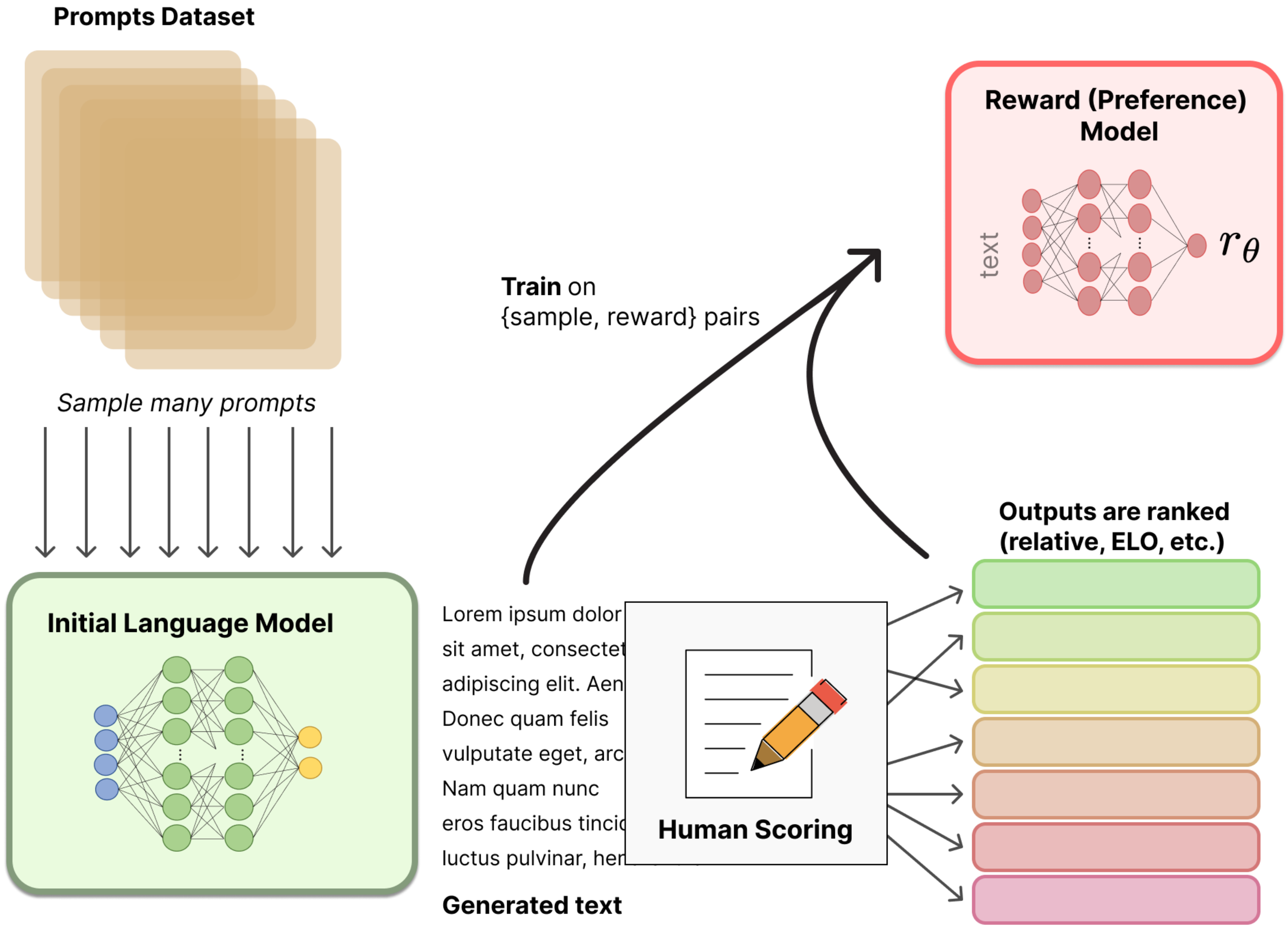
Bradly-Terry Model

- (x, y_w, y_l), where $y_w \succcurlyeq y_l, y_w, y_l \sim \pi_\theta(\,.\, | x)$
- $\mathbb{P}(y_w > y_l | x) = \dfrac{\exp R_\phi(x, y_w)}{\exp R_\phi(x, y_w) + \exp R_\phi(x, y_l)} = \sigma(R_\phi(x, y_w) - R_\phi(x, y_l))$

Preference Modelling

- (x, y_w, y_l), where $y_w \succcurlyeq y_l$,
  $y_w, y_l \sim \pi_\theta(. \mid x)$

- $L = -\log \sigma(R_\phi(x, y_w) - R_\phi(x, y_l))$



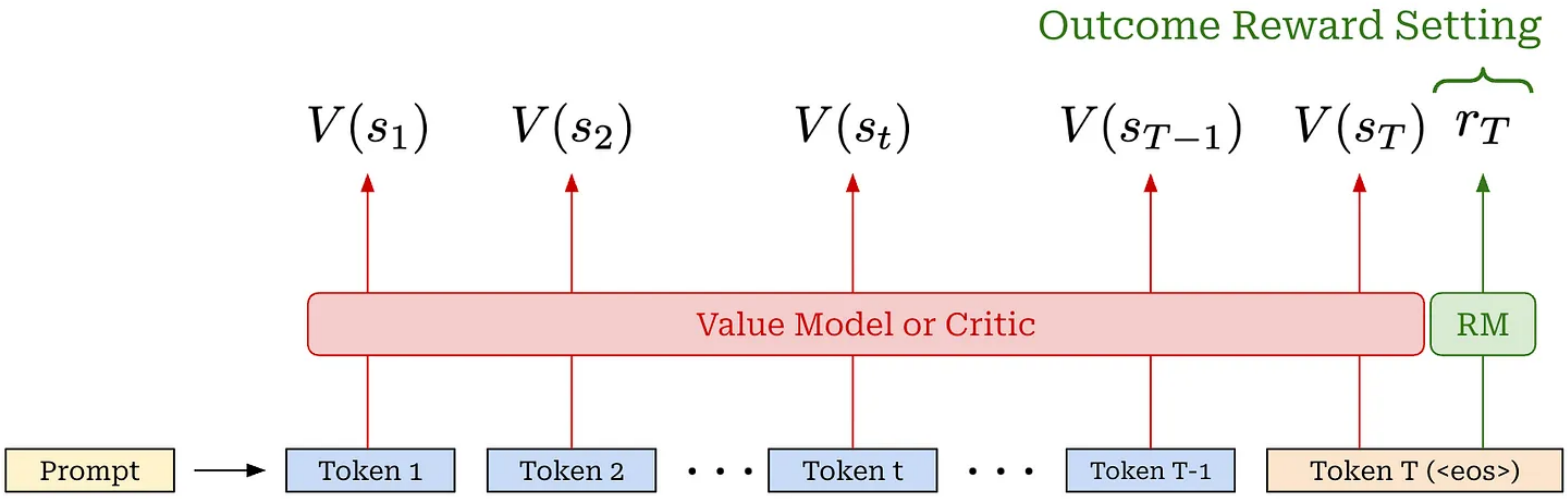
[Source](#)

RLHF Objective

- $\max_{\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta}[R_\phi(x, y) - \beta D_{KL}(\pi_\theta || \pi_{ref})]$
- RLHF improves responses but keeps the model close to the original SFT policy to avoid overoptimisation.
- KL constraint also prevents catastrophic forgetting.

LLM Critic

- Unlike a reward model $R_\phi(x, y) = r_T$, the critic is trained alongside the LLM in each policy update to ensure that its predictions remain on-policy.



[Source](#)

RLHF with PPO

- Actor objective:

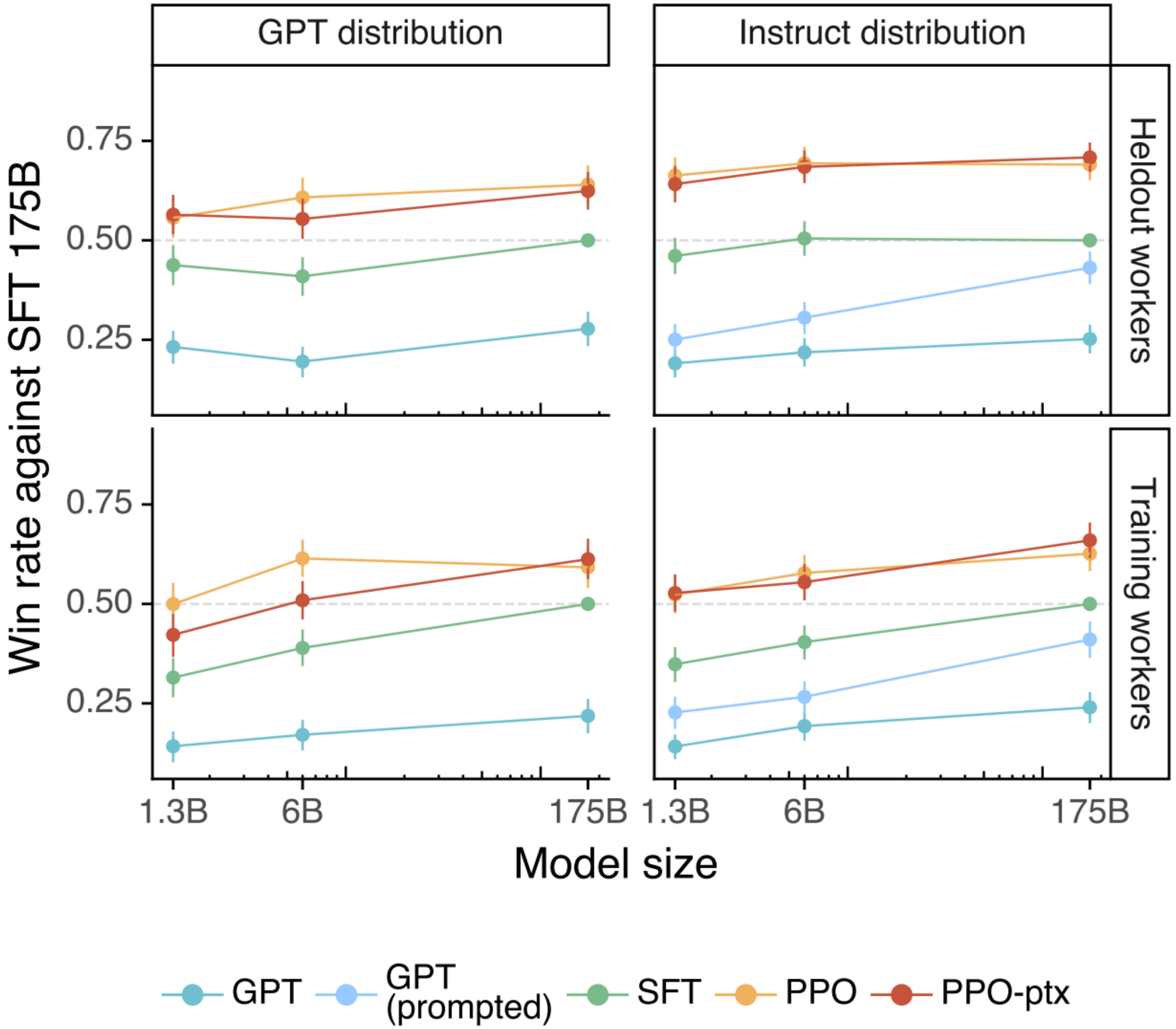
$$\mathbb{E}_t \left[\min \left(\frac{\pi_\theta(y_t \mid x)}{\pi_{\theta_{old}}(y_t \mid x)} \hat{A}_t,\ clip \left(\frac{\pi_\theta(y_t \mid x)}{\pi_{\theta_{old}}(y_t \mid x)}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) - \beta \log \frac{\pi_\theta(y_t \mid x)}{\pi_{ref}(y_t \mid x)} \right]$$

- $r_t = R_\phi(x, y) \cdot \mathbf{1}[t = T] - \beta \log \dfrac{\pi_\theta(y_t \mid x)}{\pi_{ref}(y_t \mid x)}$

- Critic objective:
- $\mathbb{E}_t \left[\left(V_\psi(x, y_{\leq t}) - G_t \right)^2 \right], G_t = \sum_{k \geq 0} \gamma^k r_{t+k}$
- $\hat{A}_t = \sum_{k=0}^{T-t} (\gamma\lambda)^k \delta_{t+k}, \qquad \delta_{t+k} = r_{t+k} + \gamma V_\psi(y_{\leq t+k+1}) - V_\psi(y_{\leq t+k})$

Results from [InstructGPT]

ChatGPT is a sibling model to InstructGPT, which is trained to follow an instruction in a prompt and provide a detailed response.



Group Relative Policy Optimization

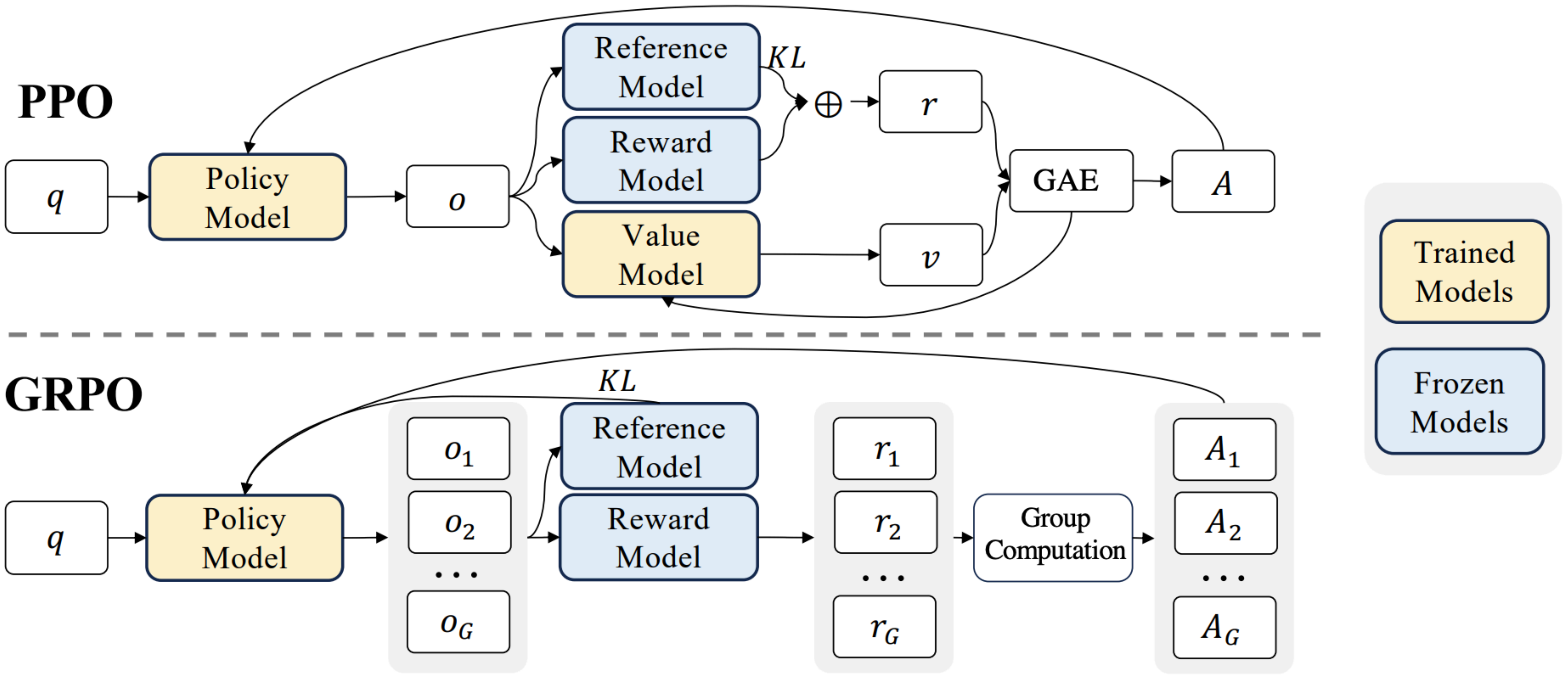
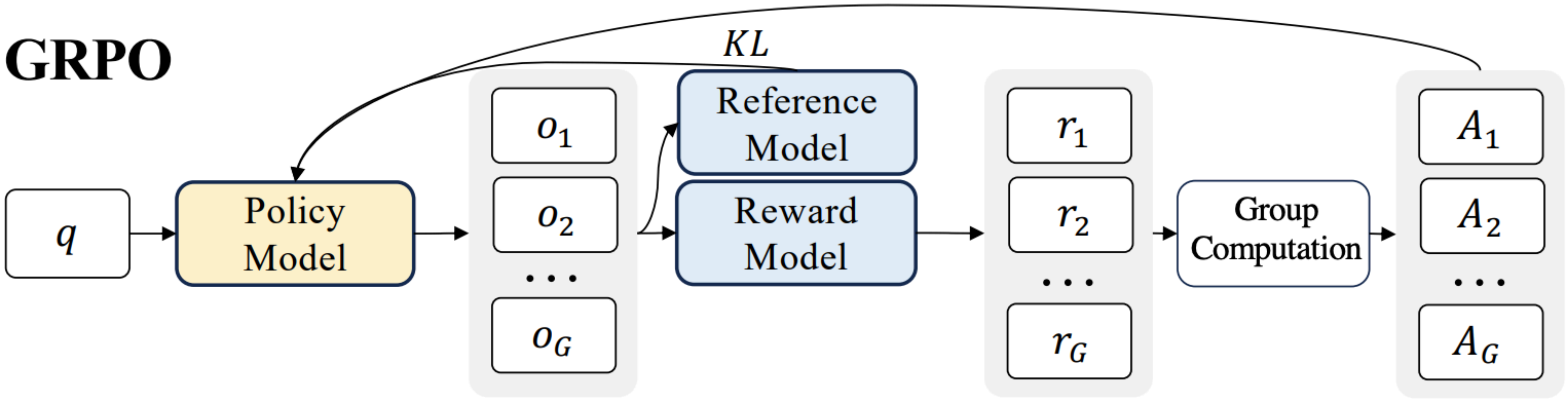


Figure 4 | Demonstration of PPO and our GRPO. GRPO foregoes the value model, instead estimating the baseline from group scores, significantly reducing training resources.

Group Relative Policy Optimization



- $\hat{A}_i = \dfrac{r_i - mean(\{r_1, \ldots, r_G\})}{std(\{r_1, \ldots, r_G\})}$

- $\mathcal{J}_{GRPO}(\theta) = \mathbb{E}_{\left[x \sim \mathcal{D}, \{y_i\}_{i=1}^{G} \sim \pi_{\theta_{old}}(\cdot|x)\right]} \dfrac{1}{G} \sum_{i=1}^{G} \dfrac{1}{|y_i|} \sum_{t=1}^{|y_i|} \left[\min \left(\dfrac{\pi_\theta(y_{i,t} \mid x, y_{i,<t})}{\pi_{\theta_{old}}(y_{i,t} \mid x, y_{i,<t})} \hat{A}_{i,t}, \text{clip} \left(\dfrac{\pi_\theta(y_{i,t} \mid x, y_{i,<t})}{\pi_{\theta_{old}}(y_{i,t} \mid x, y_{i,<t})}, 1-\varepsilon, 1+\varepsilon \right) \hat{A}_{i,t} \right) - \beta \, \mathbb{D}_{KL}\left(\pi_\theta \| \pi_{ref}\right) \right]$


prompts
completions
rewards
advantages
RM
r - mean
std
RM
r - mean
std
-
mean
objective
Policy
Ref policy
D_{KL}
D_{KL}
=

Source

GRPO vs PPO

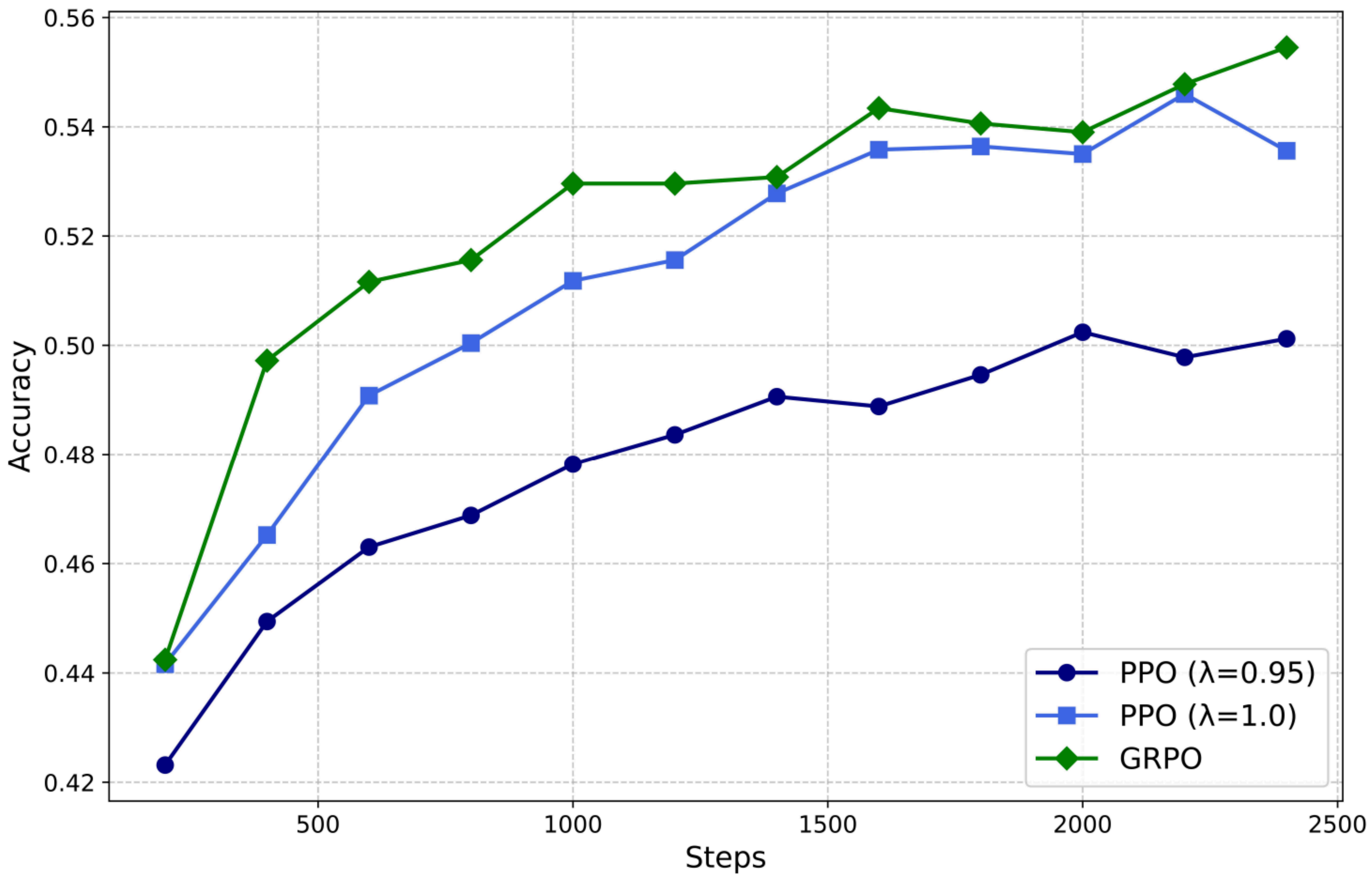
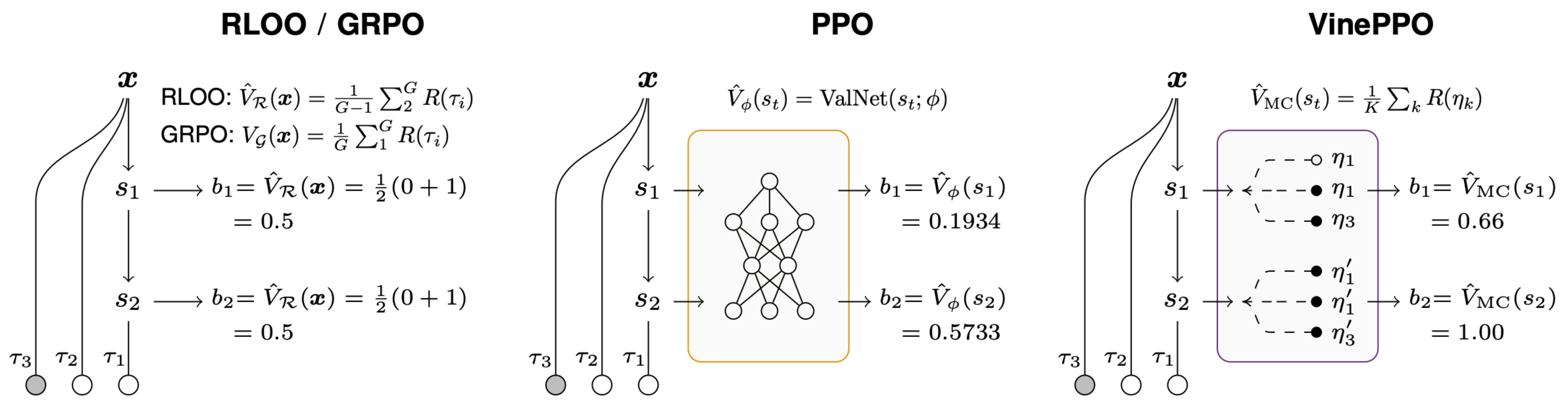


Figure 4 | Performance of PPO and GRPO on the MATH task.

VinePPO: Refining Credit Assignment in RL Training of LLMs



Direct Preference Optimization

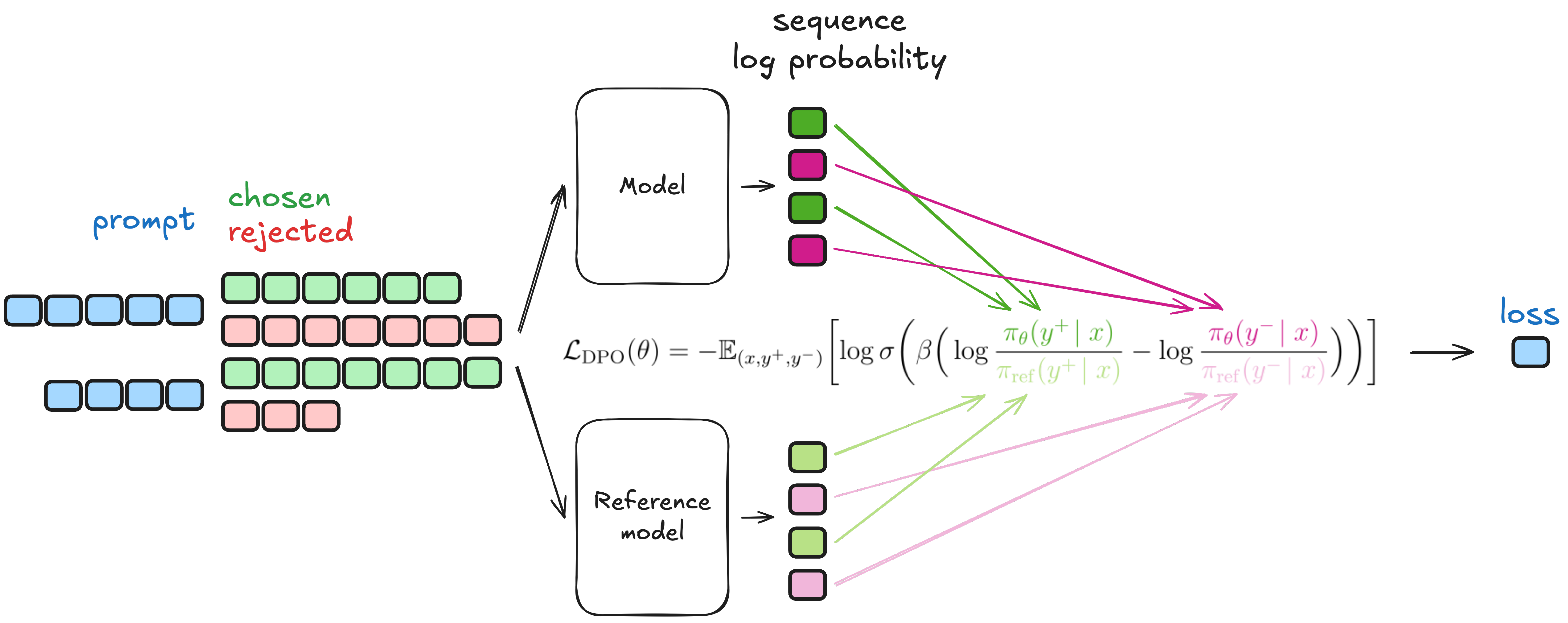
- $\max_{\pi} \mathbb{E}_{x\sim\mathcal{D},y\sim\pi_\theta}[r(x,y) - \beta D_{KL}(\pi_\theta || \pi_{ref})]$
- $\max_{\pi} \mathbb{E}_{x\sim\mathcal{D},y\sim\pi_\theta}[r(x,y) - \beta \log \frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)}]$
- $\min_{\pi} \mathbb{E}_{x\sim\mathcal{D},y\sim\pi_\theta}[\log \frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)} - \log \exp(\frac{1}{\beta}r(x,y))]$
- $\min_{\pi} \mathbb{E}_{x\sim\mathcal{D},y\sim\pi_\theta}[\log \frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)\exp(\frac{1}{\beta}r(x,y))}]$
- $\min_{\pi} \mathbb{E}_{x\sim\mathcal{D},y\sim\pi_\theta}[\log \frac{\pi_\theta(y|x)}{\frac{1}{Z(x)}\pi_{ref}(y|x)\exp(\frac{1}{\beta}r(x,y))} - \log Z(x)]$, where

$$Z(x) = \sum_y \pi_{ref}(y|x)\exp(\frac{1}{\beta}r(x,y))$$

Direct Preference Optimization

- $\pi^*(y|x) = \frac{1}{Z(x)}\pi_{ref}(y|x)\exp(\frac{1}{\beta}r(x,y))$
- $r(x,y) = \beta \log \frac{\pi^*(y|x)}{\pi_{ref}(y|x)} + \beta \log Z(x)$
- $\mathbb{P}(y_w > y_l|x) = \sigma(r(x,y_w) - r(x,y_l)) = \sigma(\beta \log \frac{\pi^*(y_w|x)}{\pi_{ref}(y_w|x)} - \beta \log \frac{\pi^*(y_l|x)}{\pi_{ref}(y_l|x)})$

- $L_{DPO} = -\mathbb{E}_{(x,y_w,y_l)\sim\mathscr{D}} \log \sigma(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{ref}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{ref}(y_l|x)}) \to \min_\theta$


sequence
log probability
prompt
chosen
rejected
Model
Reference
model
$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x,y^+,y^-)}\left[\log\sigma\left(\beta\left(\log\frac{\pi_\theta(y^+\mid x)}{\pi_{\text{ref}}(y^+\mid x)} - \log\frac{\pi_\theta(y^-\mid x)}{\pi_{\text{ref}}(y^-\mid x)}\right)\right)\right]$
loss

Source

SimPO

- $$L_{SimPO} = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}} \log \sigma\left(\frac{\beta}{|y_w|}\log \pi_\theta(y_w|x) - \frac{\beta}{|y_l|}\log \pi_\theta(y_l|x) - \gamma\right) \to \min_\theta$$

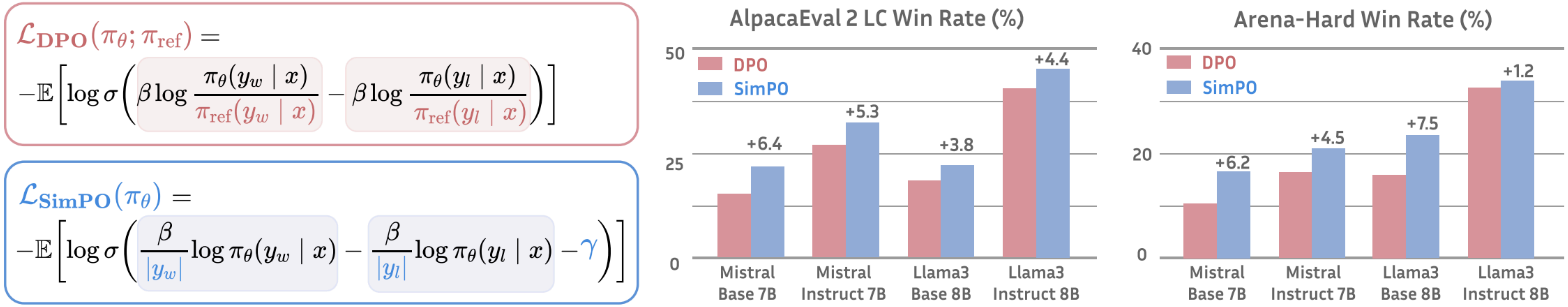


Figure 1: SimPO and DPO mainly differ in their reward formulation, as indicated in the shaded box. SimPO outperforms DPO significantly across a range of settings on AlpacaEval 2 and Arena-Hard.

Reinforcement Learning from Verifiable Reward

RLVR Objective

- $\max_{\theta} E_{x \sim \mathcal{D}, y \sim \pi_\theta}[R_{verifier}(x, y)]$
- Outcome reward which is granted after the whole completion is generated and verified.
- RLVR trains the LLM using RL with rewards derived from rule-based or deterministic verifiers.

Reasoning Model

“A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within <think> </think> and <answer> </answer> tags, respectively.”

Example

Math — GSM8K style problem

"Janet has 24 apples. She gives a third to her brother and half of the remainder to her sister. How many does she keep?"

The model generates:

<think>
A third of 24 is 8. Remainder is 24 - 8 = 16.
Half of 16 is 8. Janet keeps 16 - 8 = 8.
</think>
\boxed{8}

Verifier extracts 8 from \boxed{} and compares to the correct answer.

Example

Coding — LeetCode style problem

"Given a list of integers, return the two indices that sum to a target."

Verifier = tests execution feedback

assert solution([2, 7, 11, 15], 9) == [0, 1]
assert solution([3, 2, 4],   6) == [1, 2]
assert solution([3, 3],   6) == [0, 1]

Learning to reason via RL

If we look at some examples of reasoning trajectories from open or closed reasoning models, we will notice that these models exhibit some sophisticated reasoning behaviors in their long CoT:

- Thinking through each part of a complex problem.
- Decomposing complex problems into smaller, solvable parts.
- Critiquing solutions and finding errors.
- Exploring many alternative solutions.

Aha-Moment

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$

$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$

Rearrange to isolate the inner square root term:

$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$

...

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$

We started with the equation:

$\sqrt{a - \sqrt{a + x}} = x$

First, let's square both sides:

$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$

Next, I could square both sides again, treating the equation: $\cdots$

...

So-called aha moment detected while training RLVR with GRPO from the **base** model.

DAPO

When running the vanilla GRPO algorithm, authors notice several issues in the training process, including:

- ***Entropy collapse***: the entropy of the model's next token distribution collapses during the training process.
- ***Reward noise***: the training reward is very noisy and does not steadily increase during the RL training process.
- ***Training instability***: the training process is unstable and may diverge. We do not observe a steady increase in response length during training.

DAPO

1. Clip higher
2. Dynamic Sampling
3. Token-level loss
4. Overlong reward shaping

$$\begin{aligned}
\mathcal{J}_{\mathrm{DAPO}}(\theta) = \quad & \mathbb{E}_{(q,a)\sim\mathcal{D},\{o_i\}_{i=1}^{G}\sim\pi_{\theta_{\mathrm{old}}}(\cdot|q)} \\
& \left[\frac{1}{\sum_{i=1}^{G}|o_i|}\sum_{i=1}^{G}\sum_{t=1}^{|o_i|}\min\Big(r_{i,t}(\theta)\hat{A}_{i,t},\ \mathrm{clip}\Big(r_{i,t}(\theta),1-\varepsilon_{\mathrm{low}},1+\varepsilon_{\mathrm{high}}\Big)\hat{A}_{i,t}\Big)\right] \\
\text{s.t.} \quad & 0 < \Big|\{o_i \mid \texttt{is_equivalent}(a,o_i)\}\Big| < G.
\end{aligned}$$

Algorithm 1 **DAPO**: **D**ecoupled Clip and **D**ynamic s**A**mpling **P**olicy **O**ptimization

Input initial policy model π_θ; reawrd model R; task prompts $\mathcal{D}$; hyperparameters $\varepsilon_{\mathtt{low}}, \varepsilon_{\mathtt{high}}$

for step = 1,...,M **do**
  Sample a batch $\mathcal{D}_b$ from $\mathcal{D}$
  Update the old policy model $\pi_{\theta_{old}} \leftarrow \pi_\theta$
  Sample G outputs $\{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\mathrm{old}}}(\cdot|q)$ for each question $q \in \mathcal{D}_b$
  Compute rewards $\{r_i\}_{i=1}^{G}$ for each sampled output o_i by running R
  Filter out o_i and add the remaining to the dynamic sampling buffer (**Dynamic Sampling** Equation (11))
  if buffer size $n_b < N$:
    continue
  For each o_i in the buffer, compute $\hat{A}_{i,t}$ for the t-th token of o_i (Equation (9))
  for iteration = 1, ..., μ **do**
    Update the policy model π_θ by maximizing the DAPO objective (Equation (8))

Output π_θ

Dr. GRPO

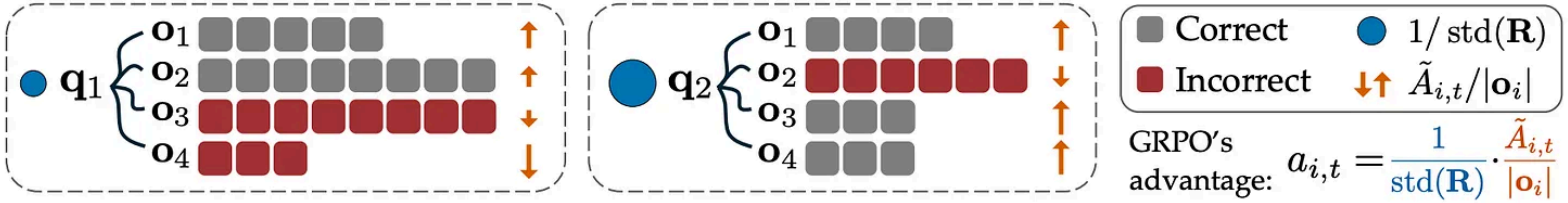


Figure 4: Illustration of the biases in GRPO. Note that the effective advantage of GRPO $a_{i,t}$ is equivalent to a reweighted version of the unbiased advantage $\tilde{A}_{i,t} = R(\mathbf{q}, \mathbf{o}_i) - \operatorname{mean}(\mathbf{R})$. The terms $\operatorname{std}(\mathbf{R})$ and $|\mathbf{o}_i|$ could bias the optimization by assigning different weights to different questions and responses, as denoted by the sizes of the blue circles and the lengths of the orange arrows. Upward arrows indicate positive advantages, and vice versa.

GRPO

$$\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|\mathbf{o}_i|}\sum_{t=1}^{|\mathbf{o}_i|}\left\{\min\left[\frac{\pi_\theta(o_{i,t}|\mathbf{q},\mathbf{o}_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|\mathbf{q},\mathbf{o}_{i,<t})}\hat{A}_{i,t}, \operatorname{clip}\left(\frac{\pi_\theta(o_{i,t}|\mathbf{q},\mathbf{o}_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|\mathbf{q},\mathbf{o}_{i,<t})}, 1-\epsilon, 1+\epsilon\right)\hat{A}_{i,t}\right]\right\},$$

$$\text{where } \hat{A}_{i,t} = \frac{R(\mathbf{q},\mathbf{o}_i) - \operatorname{mean}(\{R(\mathbf{q},\mathbf{o}_1),\ldots,R(\mathbf{q},\mathbf{o}_G)\})}{\operatorname{std}(\{R(\mathbf{q},\mathbf{o}_1),\ldots,R(\mathbf{q},\mathbf{o}_G)\})}.$$

Dr. GRPO

GRPO Done Right (without bias)

$$\frac{1}{G}\sum_{i=1}^{G}\sum_{t=1}^{|\mathbf{o}_i|}\left\{\min\left[\frac{\pi_\theta(o_{i,t}|\mathbf{q},\mathbf{o}_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|\mathbf{q},\mathbf{o}_{i,<t})}\hat{A}_{i,t}, \operatorname{clip}\left(\frac{\pi_\theta(o_{i,t}|\mathbf{q},\mathbf{o}_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|\mathbf{q},\mathbf{o}_{i,<t})}, 1-\epsilon, 1+\epsilon\right)\hat{A}_{i,t}\right]\right\},$$

$$\text{where } \hat{A}_{i,t} = R(\mathbf{q},\mathbf{o}_i) - \operatorname{mean}(\{R(\mathbf{q},\mathbf{o}_1),\ldots,R(\mathbf{q},\mathbf{o}_G)\}).$$

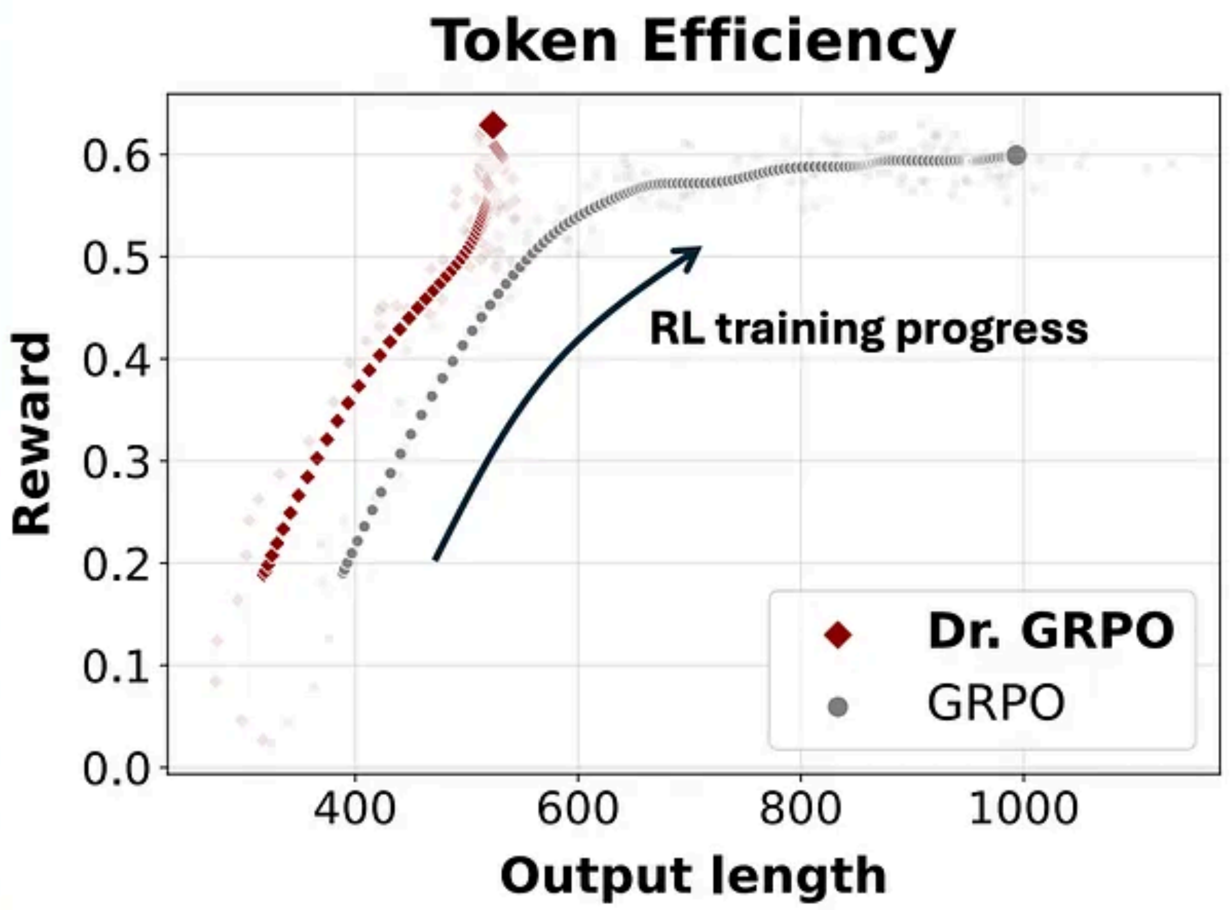


Figure 1: **Left**: Dr. GRPO introduces simple yet significant modifications to address the biases in GRPO (Shao et al., 2024), by removing the length and std normalization terms. **Right**: Our unbiased optimizer effectively prevents the model from generating progressively longer incorrect responses, thereby enhancing token efficiency.

Reinforce Leave-One-Out / CISPO

1. GRPO normalizes by the full group mean. This introduces a slight bias since the baseline is correlated with the sample being evaluated. LOO removes this bias by excluding r_i from from its own baseline.

$$\hat{A}_i^{LOO} = r_i - \frac{1}{G-1}\sum_{j \neq i} r_j$$

2. While traditional Deep-RL settings require strong regularization to reduce the high variance of the gradient estimators; this is less of a practical concern in RLHF and motivate a less computationally expensive method that preserves robustness
   - $s_t = clip\left(\frac{\pi_\theta(y_t \mid x, y_{<t})}{\pi_{\theta_{old}}(y_t \mid x, y_{<t})}, 1 - \varepsilon_{low}, 1 + \varepsilon_{high}\right)$
   - Clipped Reinforce with off-policy correction: $sg[s_t]\hat{A}_t \log \pi_\theta(y_t \mid x, y_{<t})$

CISPO

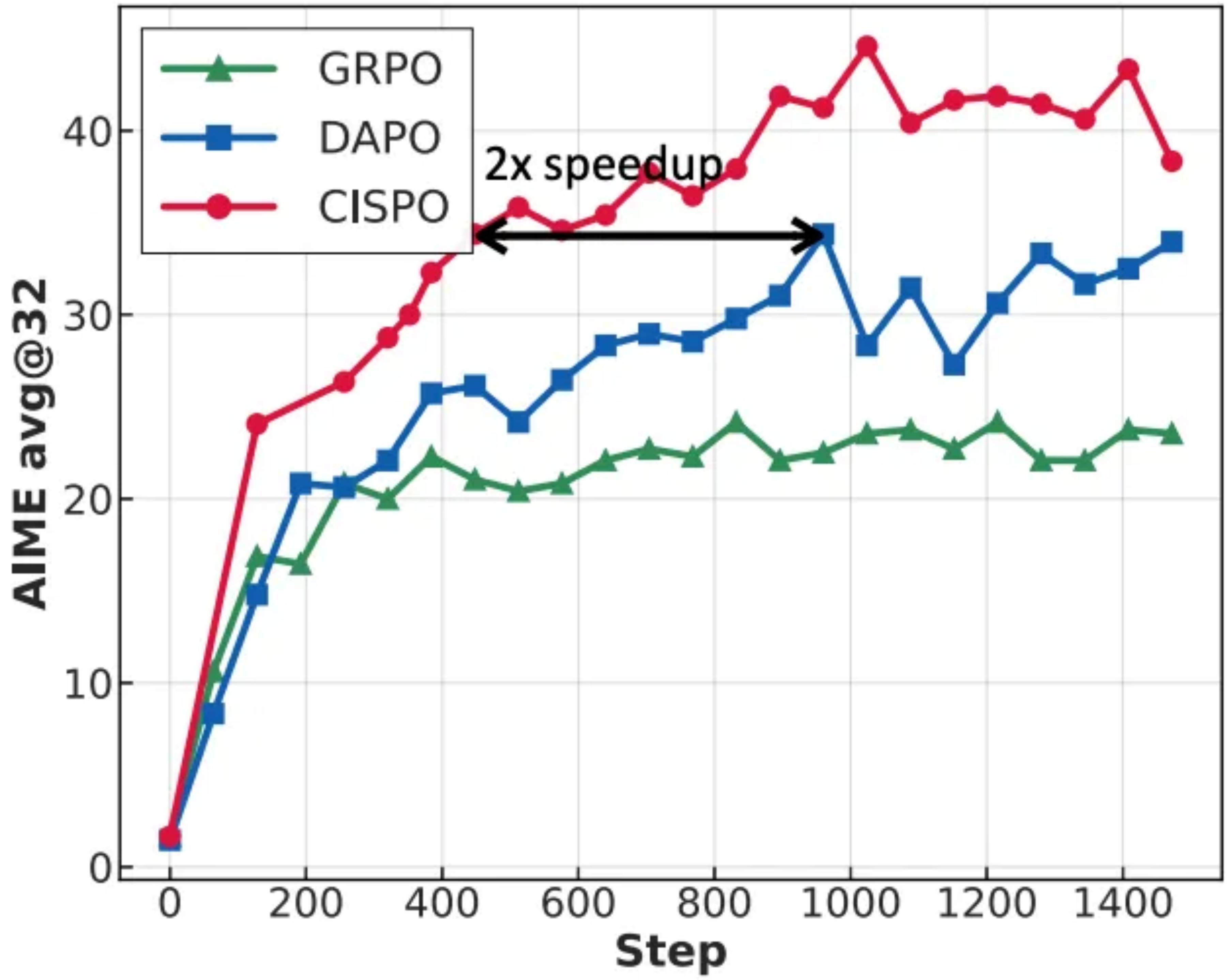
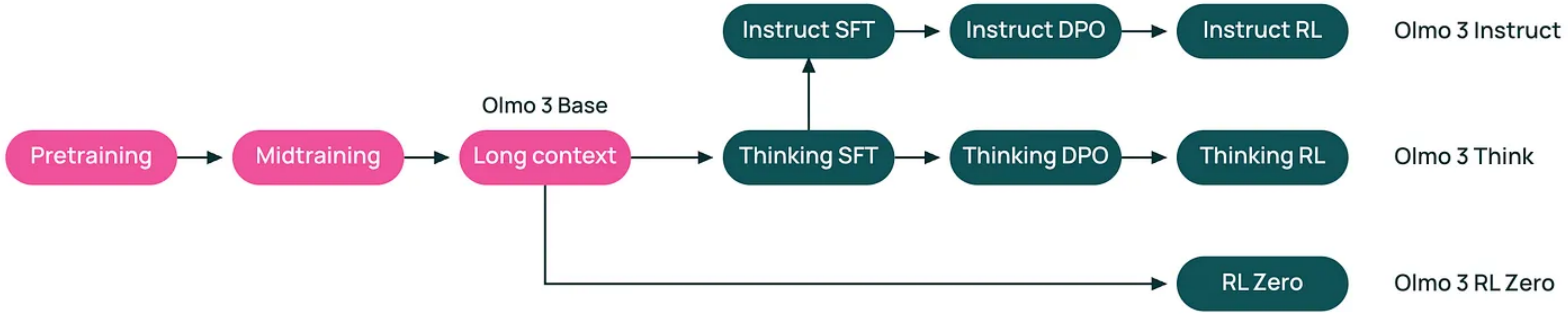


Figure 2 | Comparison of GRPO, DAPO, and our proposed CISPO on AIME 2024, based on Qwen2.5-32B-base. CISPO outperforms both GRPO and DAPO in terms of performance at the same number of training steps, and achieves comparable performance to DAPO using 50% of the training steps.

Stacking Stages



Stacking RL Stages

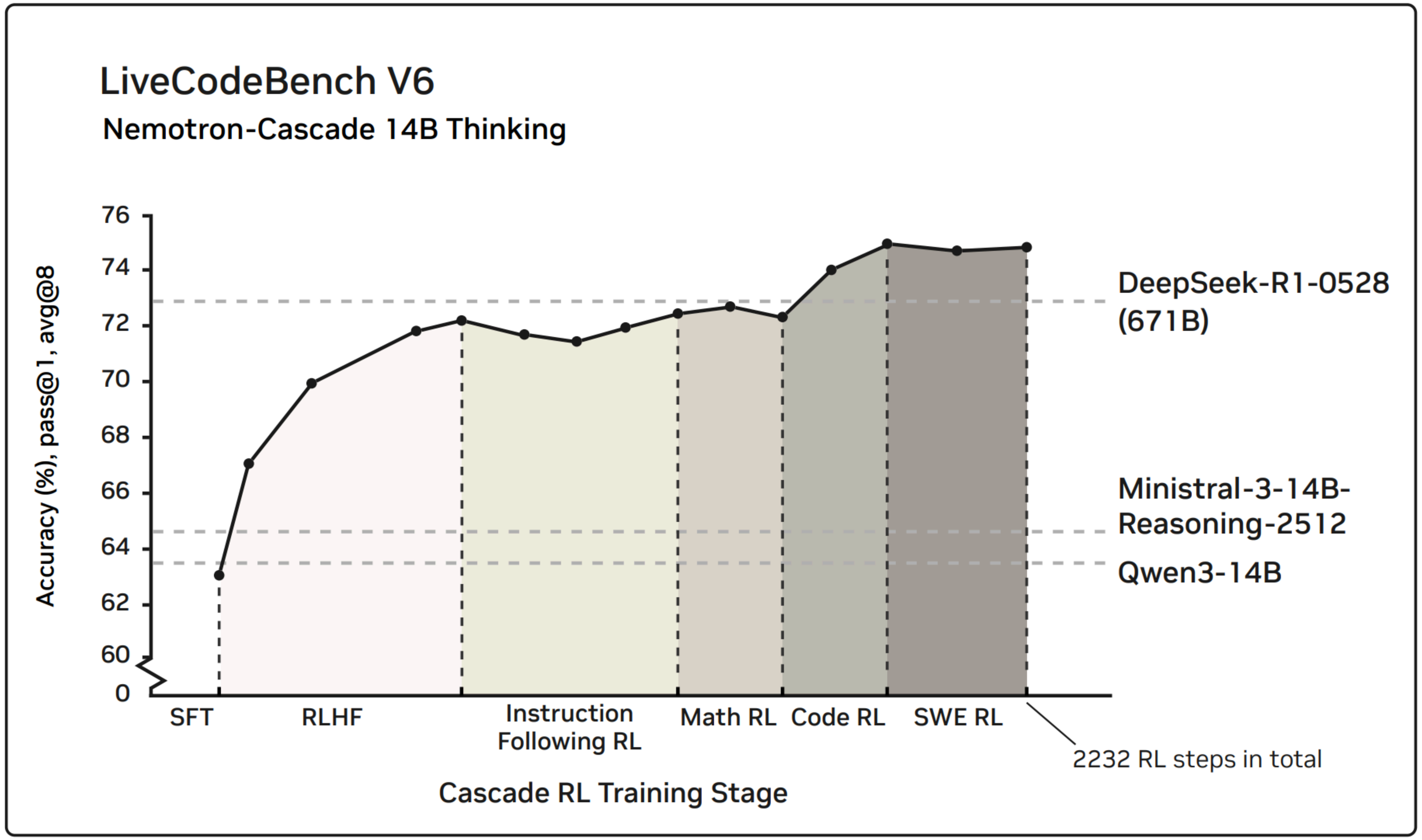


Figure 1: The LiveCodeBench v6 (08/24–05/25) performance of the Nemotron-Cascade-14B-Thinking model throughout the Cascade RL process. Note that DeepSeek-R1-0528 (671B) serves as the teacher model for SFT data curation.

Background

- Secrets of RLHF in Large Language Models Part I: PPO
- Direct Preference Optimization: Your Language Model is Secretly a Reward Model
- SimPO: Simple Preference Optimization with a Reference-Free Reward
- DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning
- GRPO++: Tricks for Making RL Actually Work
- Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?

Thank you for your attention!